

Qatar-1b

R-1937

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_131101 · created 2026-08-06T14:09:49+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456597.62965 +/- 0.00015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.023
Transit depth (Rp/Rs)^2	2.29 +/- 0.69 [%]
Orbital Inclination (inc)	82.68 +/- 0.87
Airmass coefficient 1 (a1)	0.951 +/- 0.012
Airmass coefficient 2 (a2)	0.038 +/- 0.017
Scatter in the residuals of the lightcurve fit is	1.22 %
Best Comparison Star	#3 - [521, 386]
Optimal Aperture	3.42
Optimal Annulus	14.12
Transit Duration (day)	0.06 +/- 0.0071

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.151 $\pm$ 0.023 vs 0.1463 (deviation 0.2 σ)
Duration fit vs published	0.06 d vs 0.0692 d (deviation 1.29 σ)
Mid-time O-C	0.4 min
Depth SNR	6.6
Residual scatter	1.22 %

Light curve

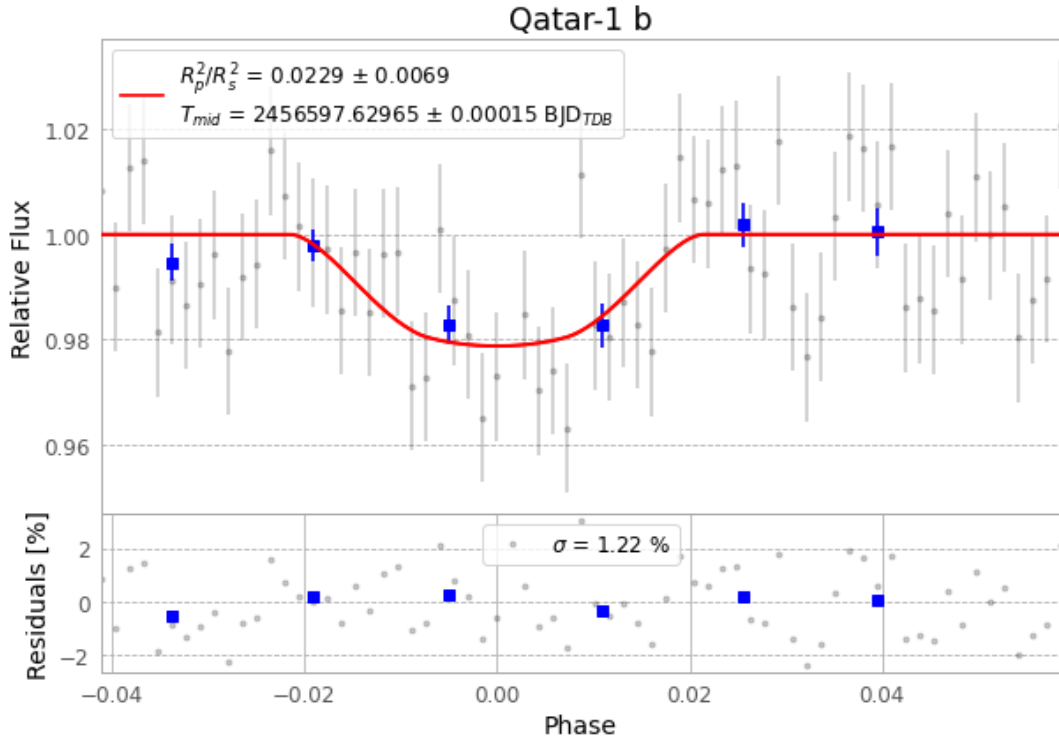


Figure 1 — FinalLightCurve_Qatar-1 b_2013-10-31.png (SHA-256 d84a54431f30af41...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [407, 308], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1627c2b4b31864b7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit d8a7143

Data provenance

69 FITS frames (45 MB), Qatar-1131101013912.FITS → Qatar-1131101050312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-131101.log (5dd1e36dc1b8...), FinalLightCurve_Qatar-1 b_2013-10-31.png (d84a54431f30...), FinalLightCurve_Qatar-1 b_2013-10-31.csv (6813f9f6c2d9...)

Compute resources

Wall time 126.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 14:09Z.